

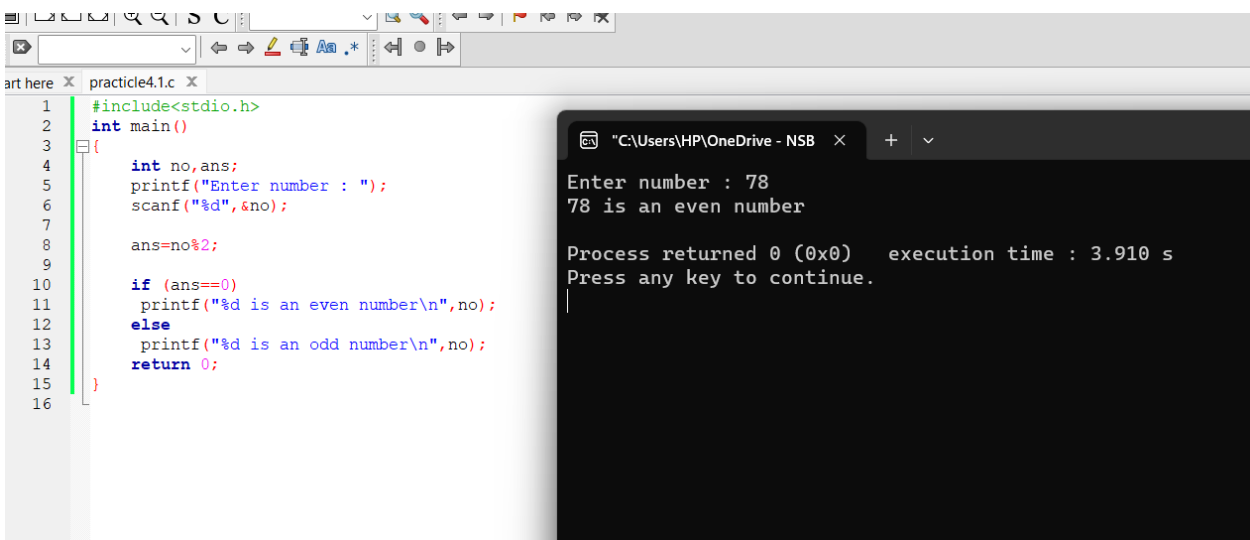
## PRACTICLE 4

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Batch No : 23.1  
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```
Q1) #include<stdio.h>
int main()
{
    int no,ans;
    printf("Enter number : ");
    scanf("%d",&no);

    ans=no%2;

    if (ans==0)
        printf("%d is an even number\n",no);
    else
        printf("%d is an odd number\n",no);
    return 0;
}
```



The screenshot displays a C program being executed. On the left, a text editor window titled 'practicle4.1.c' shows the source code. The code defines a function 'main' that prompts the user for a number, calculates its remainder when divided by 2, and prints whether it is even or odd. On the right, a command prompt window shows the execution output. The user entered '78', and the program correctly identified it as an even number. The process returned 0, and the execution time was 3.910 seconds.

```
1 #include<stdio.h>
2 int main()
3 {
4     int no,ans;
5     printf("Enter number : ");
6     scanf("%d",&no);
7
8     ans=no%2;
9
10    if (ans==0)
11        printf("%d is an even number\n",no);
12    else
13        printf("%d is an odd number\n",no);
14    return 0;
15 }
16
```

Enter number : 78  
78 is an even number  
Process returned 0 (0x0) execution time : 3.910 s  
Press any key to continue.

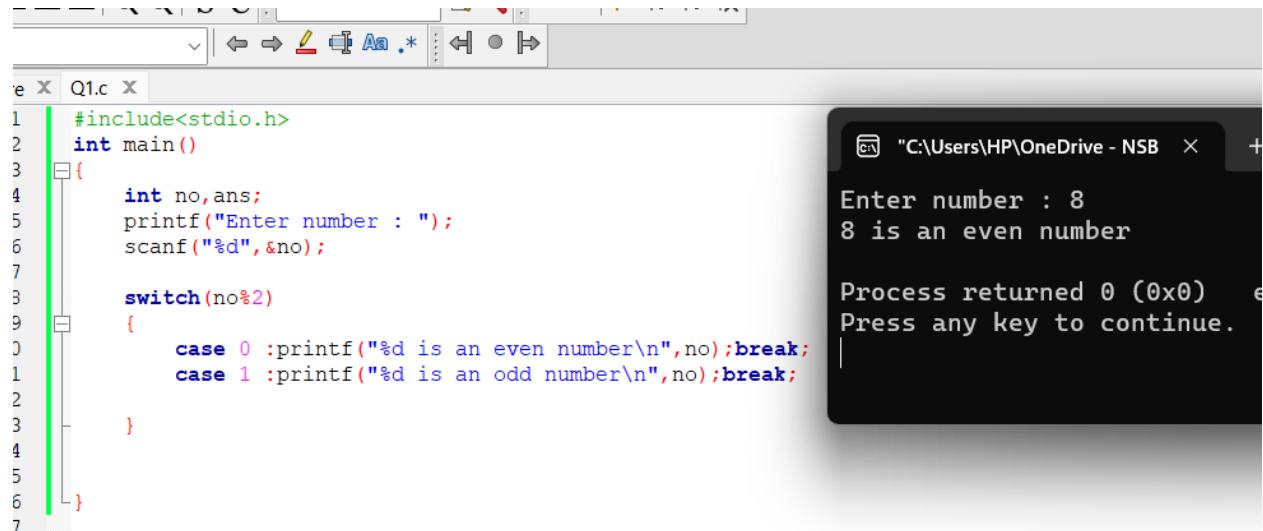
## Switch statement

```
#include<stdio.h>
int main()
{
    int no,ans;
    printf("Enter number : ");
    scanf("%d",&no);

    switch(no%2)
    {
        case 0 :printf("%d is an even number\n",no);break;
        case 1 :printf("%d is an odd number\n",no);break;

    }

}
```



The screenshot shows a C program in a text editor and its execution in a terminal. The code defines a function `main` that prompts the user to enter a number, reads it into `no`, and then uses a `switch` statement to check if `no` is even or odd based on `no % 2`. The terminal output shows that the user entered 8, and the program correctly identified it as an even number.

```
#include<stdio.h>
int main()
{
    int no,ans;
    printf("Enter number : ");
    scanf("%d",&no);

    switch(no%2)
    {
        case 0 :printf("%d is an even number\n",no);break;
        case 1 :printf("%d is an odd number\n",no);break;

    }

}
```

Enter number : 8  
8 is an even number  
Process returned 0 (0x0)  
Press any key to continue.

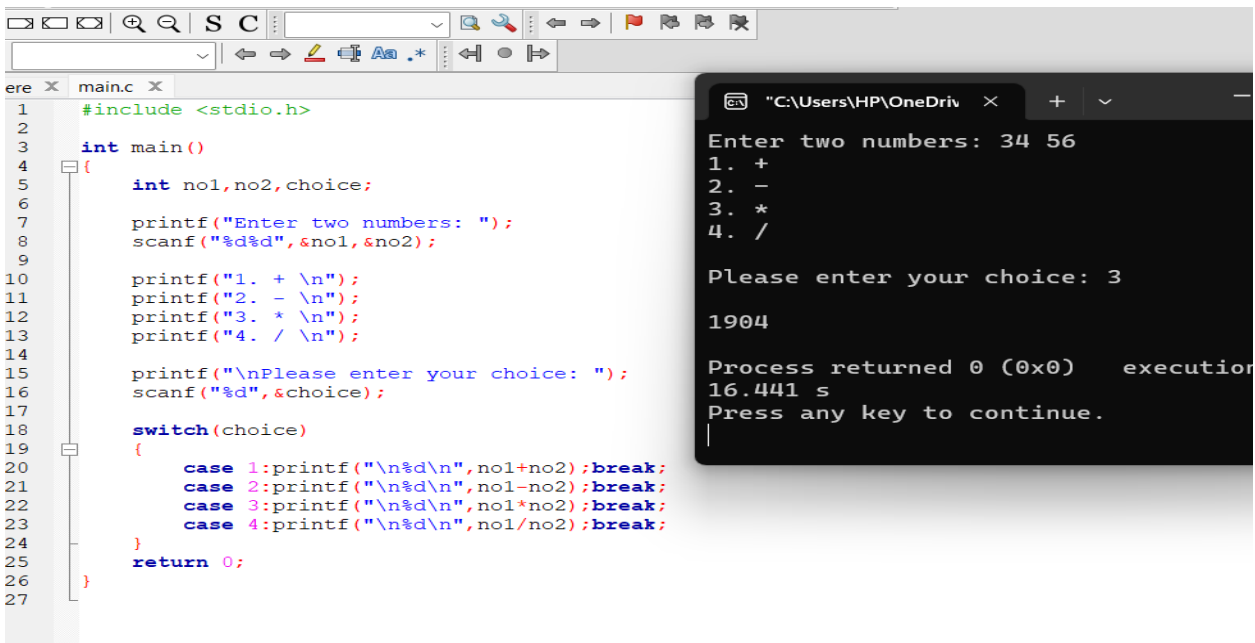
Q2)

```
#include <stdio.h>
int main()
{
    int no1,no2,choice;
    printf("Enter two numbers: ");
    scanf("%d%d",&no1,&no2);

    printf("1. + \n");
    printf("2. - \n");
    printf("3. * \n");
    printf("4. / \n");

    printf("\nPlease enter your choice: ");
    scanf("%d",&choice);

    switch(choice)
    {
        case 1:printf("\n%d\n",no1+no2);break;
        case 2:printf("\n%d\n",no1-no2);break;
        case 3:printf("\n%d\n",no1*no2);break;
        case 4:printf("\n%d\n",no1/no2);break;
    }
    return 0;
}
```



```
1 #include <stdio.h>
2
3 int main()
4 {
5     int no1,no2,choice;
6
7     printf("Enter two numbers: ");
8     scanf("%d%d",&no1,&no2);
9
10    printf("1. + \n");
11    printf("2. - \n");
12    printf("3. * \n");
13    printf("4. / \n");
14
15    printf("\nPlease enter your choice: ");
16    scanf("%d",&choice);
17
18    switch(choice)
19    {
20        case 1:printf("\n%d\n",no1+no2);break;
21        case 2:printf("\n%d\n",no1-no2);break;
22        case 3:printf("\n%d\n",no1*no2);break;
23        case 4:printf("\n%d\n",no1/no2);break;
24    }
25    return 0;
26 }
27
```

```
"C:\Users\HP\OneDrive" x + v
Enter two numbers: 34 56
1. +
2. -
3. *
4. /

Please enter your choice: 3

1904

Process returned 0 (0x0)   execution
16.441 s
Press any key to continue.
```

Q3)

```
#include <stdio.h>
int main()
{
    int choice;
    float radius, PI=3.14159;

    printf("1. Calculate the circumference of a circle\n");
    printf("2. Calculate the area of a circle\n");
    printf("3. Calculate the volume of a sphere\n");
    printf("\nEnter your choice (1-3): ");
    scanf("%d", &choice);

    printf("Enter the radius: ");
    scanf("%f", &radius);

    float circumference = 2 * PI * radius;
    float area = PI * pow(radius, 2);
    float volume = (4.0 / 3.0) * PI * pow(radius, 3);

    switch (choice)
    {
        case 1:printf("The circumference of the circle is: %.2f\n",
                     circumference);break;
        case 2:printf("The area of the circle is: %.2f\n", area);break;
        case 3:printf("The volume of the sphere is: %.2f\n",
                     volume);break;
        default:printf("Invalid choice!\n");break;
    }

    return 0;
}
```

```
1 | #include <stdio.h>
2
3 | int main()
4 | {
5 |     int choice;
6 |     float radius, PI=3.14159;
7
8 |     printf("1. Calculate the circumference of a circle\n");
9 |     printf("2. Calculate the area of a circle\n");
10 |    printf("3. Calculate the volume of a sphere\n");
11 |    printf("\nEnter your choice (1-3): ");
12 |    scanf("%d", &choice);
13
14 |    printf("Enter the radius: ");
15 |    scanf("%f", &radius);
16
17 |    float circumference = 2 * PI * radius;
18 |    float area = PI * pow(radius, 2);
19 |    float volume = (4.0 / 3.0) * PI * pow(radius, 3);
20
21 |    switch (choice)
22 |    {
23 |        case 1:printf("The circumference of the circle is: %.2f\n", circumference);break;
24 |        case 2:printf("The area of the circle is: %.2f\n", area);break;
25 |        case 3:printf("The volume of the sphere is: %.2f\n", volume);break;
26 |        default:printf("Invalid choice!\n");break;
27 |    }
28
29 |    return 0;
30 | }
31
```

```
"C:\Users\HP\OneDrive" x + - □ x
1. Calculate the circumference of a circle
2. Calculate the area of a circle
3. Calculate the volume of a sphere

Enter your choice (1-3): 2
Enter the radius: 14
The area of the circle is: 615.75

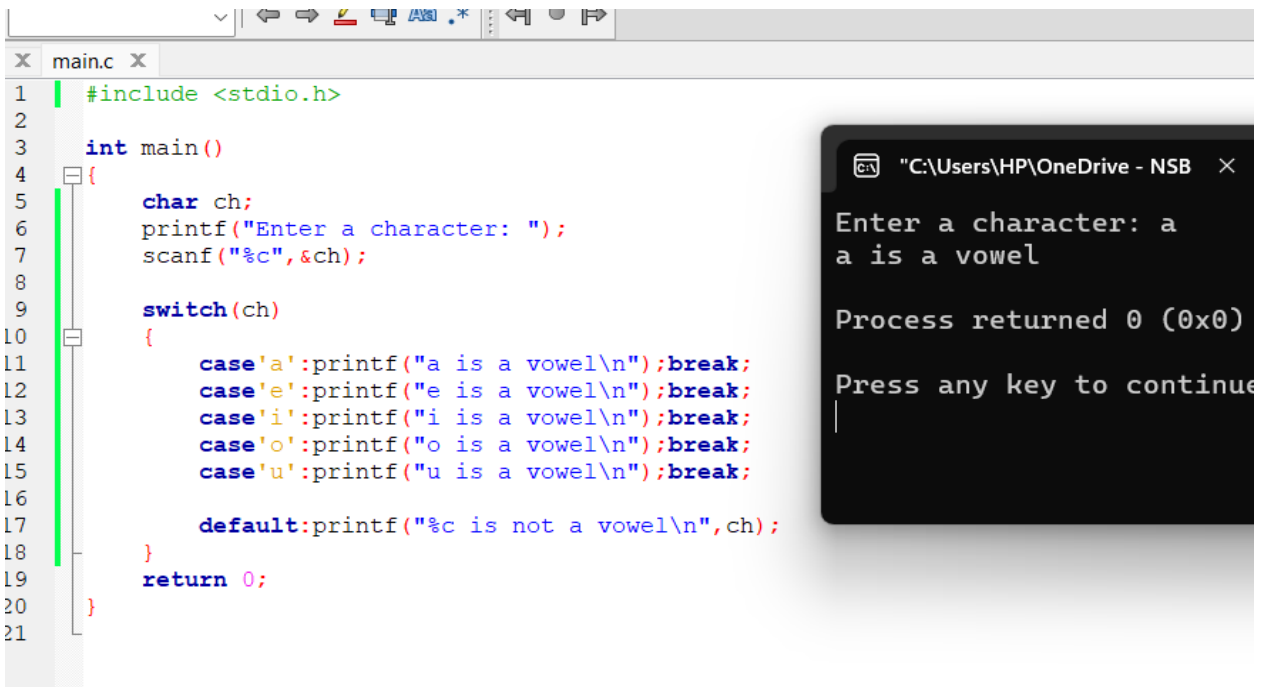
Process returned 0 (0x0)   execution time :
9.962 s
Press any key to continue.
```

Q4)

```
#include <stdio.h>
int main()
{
    char ch;
    printf("Enter a character: ");
    scanf("%c",&ch);

    switch(ch)
    {
        case'a':printf("a is a vowel\n");break;
        case'e':printf("e is a vowel\n");break;
        case'i':printf("i is a vowel\n");break;
        case'o':printf("o is a vowel\n");break;
        case'u':printf("u is a vowel\n");break;

        default:printf("%c is not a vowel\n",ch);
    }
    return 0;
}
```



```
1  #include <stdio.h>
2
3  int main()
4  {
5      char ch;
6      printf("Enter a character: ");
7      scanf("%c",&ch);
8
9      switch(ch)
10     {
11         case'a':printf("a is a vowel\n");break;
12         case'e':printf("e is a vowel\n");break;
13         case'i':printf("i is a vowel\n");break;
14         case'o':printf("o is a vowel\n");break;
15         case'u':printf("u is a vowel\n");break;
16
17         default:printf("%c is not a vowel\n",ch);
18     }
19     return 0;
20 }
21
```

"C:\Users\HP\OneDrive - NSB

Enter a character: a  
a is a vowel

Process returned 0 (0x0)

Press any key to continue

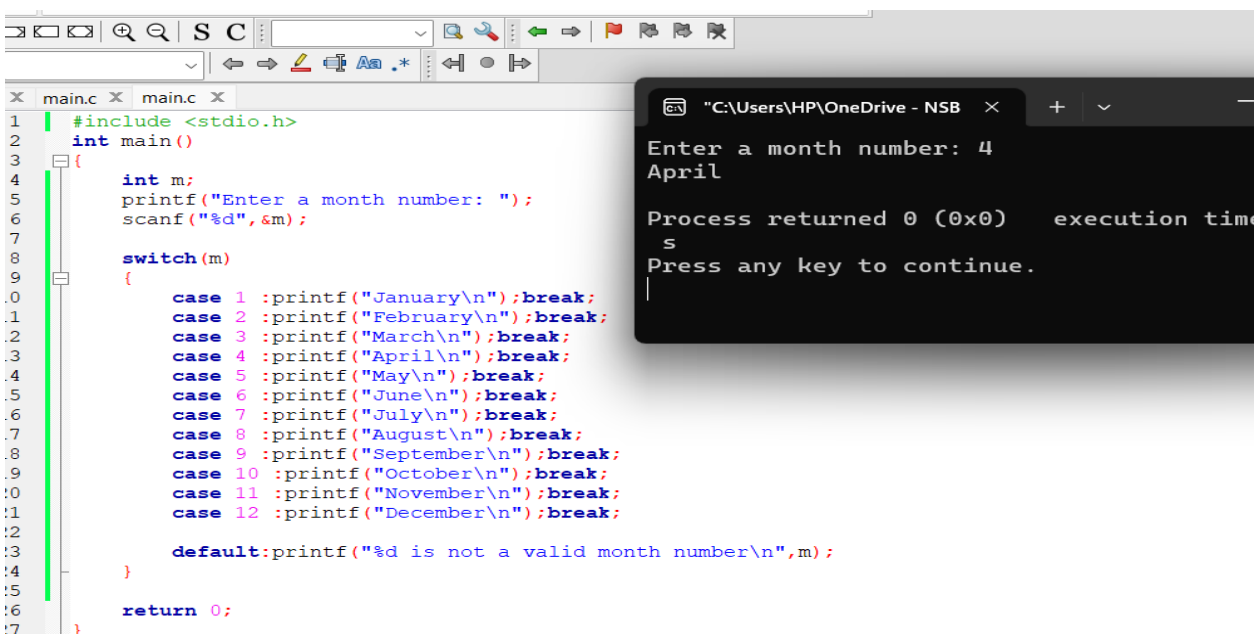
```

Q5) #include <stdio.h>
int main()
{
    int m;
    printf("Enter a month number: ");
    scanf("%d",&m);

    switch(m)
    {
        case 1 :printf("January\n");break;
        case 2 :printf("February\n");break;
        case 3 :printf("March\n");break;
        case 4 :printf("April\n");break;
        case 5 :printf("May\n");break;
        case 6 :printf("June\n");break;
        case 7 :printf("July\n");break;
        case 8 :printf("August\n");break;
        case 9 :printf("September\n");break;
        case 10:printf("October\n");break;
        case 11:printf("November\n");break;
        case 12:printf("December\n");break;

        default:printf("%d is not a valid month number\n",m);
    }
    return 0;
}

```



The screenshot shows a C program being executed. On the left, a text editor displays the source code, which is identical to the code block above. The code uses a switch statement to map month numbers 1 through 12 to their respective names. On the right, a terminal window shows the program's execution. It prompts the user to 'Enter a month number: ', the user enters '4', and the program outputs 'April'. Below this, the terminal shows 'Process returned 0 (0x0) execution time: 5' and 'Press any key to continue.'.

```

main.c x main.c x
1 | #include <stdio.h>
2 | int main()
3 | {
4 |     int m;
5 |     printf("Enter a month number: ");
6 |     scanf("%d",&m);
7 |
8 |     switch(m)
9 |     {
10 |         case 1 :printf("January\n");break;
11 |         case 2 :printf("February\n");break;
12 |         case 3 :printf("March\n");break;
13 |         case 4 :printf("April\n");break;
14 |         case 5 :printf("May\n");break;
15 |         case 6 :printf("June\n");break;
16 |         case 7 :printf("July\n");break;
17 |         case 8 :printf("August\n");break;
18 |         case 9 :printf("September\n");break;
19 |         case 10 :printf("October\n");break;
20 |         case 11 :printf("November\n");break;
21 |         case 12 :printf("December\n");break;
22 |
23 |         default:printf("%d is not a valid month number\n",m);
24 |     }
25 |
26 |     return 0;
27 | }

```

```

"C:\Users\HP\OneDrive - NSB" x + v
Enter a month number: 4
April
Process returned 0 (0x0) execution time: 5
Press any key to continue.

```